Dense Retrieval Is Degenerate Associative Memory: A Reduction and a Regime Taxonomy for Agent Long-Term Memory

Pushkal Gupta
School of Computer Science and Engineering

Rajit Agrawal
School of Computer Science and Engineering

Anshveer Singh
School of Computer Science and Engineering

Abstract

Agent long-term memory is almost universally retrieved by top-k embedding similarity. Associative-memory theory offers an alternative—recall as convergence to an attractor in an energy landscape—but it has never been evaluated as the *persistent memory layer* of an agent under matched conditions. We first show why the two literatures have not met: at the standard operating point they are the same operator. Proposition 1 establishes that dense top-k retrieval is exactly the (T=1, $\beta \to \infty$, b=0, R=I) corner of the modern Hopfield update, verified numerically on 1000/1000 queries on synthetic vectors and 838/838 on real sentence embeddings. The industry baseline is therefore a configuration of our own code rather than a separate system, which removes implementation advantage by construction and converts an unanswerable question into a sharp one: *which departures from that corner pay for themselves, and where?*

We answer it across two modalities with a single frozen encoder, thread-disjoint hyperparameter selection, and cluster-bootstrap intervals throughout: 5,847 held-out image queries, 1,977 LoCoMo questions, 146 multi-target text questions, and 10,638 temporal-association probes across two corpora. The answer is regime-dependent and mostly negative. Iterative descent beats cosine on image pattern completion (+0.014 pooled Recall@1, 95% CI $[+0.006, +0.021]$), and the advantage *grows monotonically with corruption severity*, from a null +0.003 at 30% occlusion to +0.021 at 70%. It loses monotonically on conversational question→evidence retrieval: every increment of descent costs accuracy, from 0.336 to 0.252 Coverage@5 across an interpolation sweep whose endpoint provably reproduces cosine. One mechanism explains both signs. Associative completion requires the cue to be a degraded instance of a stored pattern; image occlusion satisfies that premise and question answering violates it.

We further show that β—the only dial with leverage—is influential only inside a narrow, encoder-specific band ($\approx[32, 64]$ on the vision backbone we measure it on), above which descent degenerates into cosine — at β=512 it returns cosine's own memory on 99.4% of queries. A comparison run above that band compares cosine against itself, which we argue is a live hazard for this literature rather than a detail of ours. Of eight mechanisms tested beyond the reduction corner, exactly one survives: a nonparametric predictor over the agent's own observed transitions beats cosine on one-hop temporal association, reproduced in three independent implementations on two corpora. On LoCoMo, against a pooled bank of 5,882 turns and over 5,233 probes in 272 sessions, it raises Recall@1 from 0.064 to 0.107 (+0.043, $[+0.036, +0.050]$, $p < 10^{-4}$; +67% relative) and Recall@5 from 0.174 to 0.251. The same experiment kills the multi-hop extension: at two hops the effect is −0.002

[−0.008, +0.004], p=0.77, and we withdraw a previously recorded +0.185 two-hop result of our own as an artifact. Multi-step descent, residual peeling, repulsive parallel descent, vote-count readout, multi-scale β, per-memory β, retrieval-margin calibration, and coupling the predictor into the energy functional all fail, each with a diagnosed mechanism rather than a bare null.

Finally, we report the measured magnitude of six protocol errors we committed and corrected. Each would have yielded a publishable but false result: a non-dimensionless hyperparameter (0.016 vs. 0.277 after normalisation), a pseudo-replicated bootstrap (p=0.0005 vs. p=0.076), test-set hyperparameter selection (+0.042 vs. +0.004), asymmetric candidate exclusion (an exact 0.000 that read as a finding), an unmatched step size (−0.115 vs. −0.062), and a significant result whose script was never committed and does not reproduce (+0.049, p=0.032 vs. +0.024, p=0.24). Two of the six were false *negatives*, which proved materially harder to find. We release all code, the regime taxonomy, and an append-only experiment ledger in which every entry above appears with its date, its question, and the entries it overturned.

1 Introduction

A large language model is a function from text to text with no state between calls. An *agent* is what results when that function is asked to operate over days or months, and the forgetting becomes the binding constraint. The standard remedy is an external store: write experience to it, and on each turn retrieve a few items to place back in context. The retrieval step is the subject of this paper.

Across the agent-memory literature—MemGPT (Packer et al., 2023), MemoryBank (Zhong et al., 2024), A-MEM (Xu et al., 2025), HippoRAG (Gutiérrez et al., 2024)—the sophistication is concentrated almost entirely in *what gets written and when it is evicted*. The read operator itself has been a frozen cosine nearest-neighbour lookup for a decade. Three objections to it are standard. It gets one shot at an ambiguous cue; it weights a memory recalled four hundred times identically to one glimpsed once, patched by hand-tuned recency and usage heuristics bolted to the side of a similarity score; and it can only return what is *similar*, which is structurally the wrong relation for recall by association—lightning does not resemble thunder.

A separate literature has studied exactly this operator for forty years. Associative memory treats recall as a dynamical process: a cue is released into an energy landscape whose minima are the stored patterns, and it descends until it settles (Hopfield, 1982). Modern variants hold exponentially many patterns (Krotov & Hopfield, 2016; Ramsauer et al., 2021) and repair a corrupted cue as a side effect of converging. The obvious experiment—use one as an agent's persistent memory and compare—has not been run under matched conditions.

Why not. We think the answer is Proposition 1, and that it is the reason this comparison has been quietly circular wherever it has been attempted. At the operating point everyone actually uses, the two methods are the *same operator*. One step of the modern Hopfield update with the temperature turned up returns the top cosine neighbour exactly. Dense retrieval is not a rival sitting outside the energy formulation; it is a single corner of it. Any paper claiming that an energy-based memory beats similarity search must therefore say which coordinate it moved, and by how far—and if it ran at high β, it compared cosine against itself.

Stating the reduction first changes the question from *"is energy-based memory better?"*, which is not well posed, to *"which departures from the dense corner pay for themselves, and in which regime?"*, which is. It also makes every comparison in this paper exact by construction: the baseline is our own code with different arguments, so there is no implementation advantage to argue about and no encoder mismatch to suspect.

What we found. Mostly that the departures do not pay, and—more usefully—why. Of the mechanisms available beyond the corner, every one that changes only *how an answer is read off the landscape* fails on agent text: multi-step descent, residual peeling, repulsive parallel descent, vote-count readout, multi-scale and per-memory β. The one mechanism that helps does not touch the landscape at all; it is a model of *what follows what*, fitted on the agent's own transitions.

Between modalities the sign of the effect flips, and it flips for a reason we can state in one sentence and verify in both directions. Energy descent computes a weighted average of stored patterns and moves the retrieval state onto their span. That pays off exactly when the query is a degraded copy of something already stored, because moving toward the memory manifold restores lost signal; it costs accuracy when the query is not, because averaging smears the discriminative content of the cue. Image occlusion satisfies the premise and descent wins by more as damage increases. Question→evidence retrieval violates it and descent loses monotonically. Both halves are measured, not argued.

Contributions.

- **C1 The reduction.** Dense top-k retrieval is exactly the (T=1, $\beta \to \infty$, b=0, R=I) corner of the modern Hopfield update (Proposition 1), verified at 1000/1000 queries on synthetic vectors and 838/838 on real sentence embeddings, both asserted in continuous integration. Every baseline in this paper is a configuration of the method under test. (X1)

- **C2 The β leverage band, and a hazard it implies.** β has influence only within a narrow band, bounded below by collapse of the state to the corpus centroid on the first step and above by degeneration into cosine. Accuracy cannot see the upper edge—two retrievers can score identically while disagreeing about what they return—so we measure top-1 *decision* agreement with cosine, which rises to **99.4**% at β=512. A comparison run above the band therefore compares cosine against itself on nearly every query and will correctly report no difference for the wrong reason. The band is a property of the encoder, not a constant. (X3, X11, §3.4)

- **C3 A regime taxonomy with a falsifiable premise.** Completion, composition and semantic match require mutually exclusive operating points, and which one a query belongs to is predictable from whether the cue is a degraded instance of a stored pattern. Evidence in both directions with dose-response: +0.014 pooled [+0.006, +0.021] on held-out image completion rising to +0.021 at 70% occlusion, against a monotone −0.084 across an interpolation sweep on 1,977 LoCoMo questions. (§6)

- **C4 Eight diagnosed negative results.** Each is reported with the mechanism that produces it rather than as a bare null, including two that overturn claims we had previously made ourselves. (§5.4)

- **C5 One externally validated positive, and a sharp limit on it.** A nonparametric predictor over the agent's own observed transitions beats cosine on one-hop temporal association, in three independent implementations on two corpora. On the public LoCoMo benchmark it raises Recall@1 from 0.064 to 0.107 (+0.043, [+0.036, +0.050], $p < 10^{-4}$, +67% relative) over 5,233 probes in 272 sessions against a pooled 5,882-turn bank. At *two* hops the same experiment returns −0.002 [-0.008, +0.004]: association does not compose, and we withdraw a +0.185 two-hop figure of our own. (§5.3, X26)

- **C6 The measured price of six protocol shortcuts.** For each of six errors we made, the magnitude of the false result it produced in our own data. Two were false *negatives*, which proved far harder to find than the false positives. (§7)

What this paper is not. It is not a system paper, and it does not claim a better agent memory. The thesis we began with—that energy-based retrieval improves agent recall—is refuted here by our own measurements, at adequate power, with the mechanism identified. We think the resulting object is more useful to the field than the system would have been: a reduction that makes the comparison exact, a taxonomy that predicts the sign of the effect before the experiment is run, and a quantified account of how easy it is to publish the opposite conclusion.

2 Related Work

2.1 Agent memory systems: sophistication on the write path

MemGPT frames long-term memory as an operating-system paging problem, with the model managing transfers between a context window and external storage (Packer et al., 2023). MemoryBank adds a forgetting curve over stored summaries (Zhong et al., 2024). A-MEM constructs interlinked notes with agent-generated structure (Xu et al., 2025). HeLa-Mem introduces Hebbian co-activation and an episodic-to-semantic distillation stage (Zhu et al., 2026). These systems differ substantially in *what is written, summarised, linked and evicted*, and they are near-identical in how a memory is read: embed the query, take the top-k cosine neighbours.

HippoRAG is the most prominent attempt to move past that read operator, replacing flat similarity with personalised PageRank over a constructed knowledge graph (Gutiérrez et al., 2024). Its association structure is *spatial*—built from extracted entities and relations—where ours is *temporal*, learned from the order in which the agent's own episodes arrived, with no extraction or graph-construction stage. The two are complementary rather than competing, and we do not claim superiority over it; we have not run the comparison.

2.2 Associative and energy-based memory: sophistication on the read path

The classical Hopfield network stores patterns as minima of an energy function and recalls by descent (Hopfield, 1982), with capacity $\approx 0.14d$ before basins merge. Dense associative memory raises capacity with higher-order interactions (Krotov & Hopfield, 2016), and the continuous modern formulation attains exponential capacity with a one-step retrieval guarantee (Ramsauer et al., 2021). Universal Hopfield Networks place these variants in one framework of similarity, separation and projection (Millidge et al., 2022). Dynamic Manifold Hopfield Networks modulate attractor geometry by context (Li et al., 2025), and coupled-oscillator formulations separate stored patterns from a learned routing layer (Guo & Pasqualetti, 2026)—the latter directly motivating our decision to keep R separate from X in Equation (2).

This literature almost exclusively evaluates on synthetic patterns or image reconstruction. Our completion result (§5.1) is consistent with it, and §5.2 is the finding that does not transfer: the regime these papers study is the one in which the mechanism works, and agent memory retrieval is predominantly not that regime.

2.3 The equivalence, and why it has not been exploited

Ramsauer et al. (2021) prove that the modern Hopfield update is the attention mechanism of the transformer. This is usually read as a statement about transformers. Read in the other direction it says that a one-step, high-β Hopfield retrieval over a memory bank *is* an attention read over that bank—and, with normalised patterns and no strength term, is exactly cosine top-k. Proposition 1 is a corollary of their result rather than a new theorem, and we claim no priority for it. What we claim is the consequence: it is the reason the comparison between the two literatures has been circular wherever it has been attempted, and it should be stated as the first line of any such comparison rather than discovered in review.

2.4 Prediction in representation space

The joint-embedding predictive architecture proposes predicting the representation of unobserved content rather than the content itself (LeCun, 2022; Assran et al., 2023). Our module P (§3.5) is a JEPA-shaped predictor over an episode stream, and §5.3 is, to our knowledge, the first report that the *parametric* version of this fails in the low-data regime an individual agent's history actually provides (1.000 on fitted storylines, 0.000 on unseen), with a diagnosis—near-orthogonal displacements, no global transition direction—rather than a tuning complaint.

Predictive Associative Memory is the closest prior work: it retrieves by temporal co-occurrence using an external predictor alongside the memory (Dury, 2026). Our original claimed contribution against it was *integration*—folding the predictor into the energy functional instead of running it beside. §5.4 reports that

this does not work, at any coupling strength, and withdraws the claim. What remains distinct is the reduction frame, the regime taxonomy, and the nonparametric form of the predictor.

2.5 Mixtures, sampling, and statistical practice

The rank argument for mixtures of softmaxes (Yang et al., 2018) establishes that our integrated multi-β variant is not reducible to any single β; we confirm the expressiveness empirically and find it performance-neutral (Table 6). The stochastic relaxation we specify and then decline to build on rests on Boltzmann machines (Ackley et al., 1985) and Langevin sampling (Welling & Teh, 2011). Residual peeling is matching pursuit (Mallat & Zhang, 1993) over the energy landscape, and repulsive parallel descent is Stein-variational in flavour (Liu & Wang, 2016). Our interval procedure follows the clustered bootstrap (Cameron et al., 2008); the headroom that makes any of these measurable at all on text is a consequence of embedding anisotropy (Ethayarajh, 2019; Gao et al., 2019), without which, as §4 notes, cosine saturates and nothing can be compared.

3 The Reduction

3.1 Setup and notation

Memories $m_1, \dots, m_N$ are embedded by a frozen encoder E into unit vectors $z_i = E(m_i) \in \mathcal{S}^{d-1}$ and stacked as $X = [z_1 \cdots z_N] \in \mathbb{R}^{d \times N}$. A cue x is embedded to $q = E(x)$, and the retrieval state is initialised at $\boldsymbol{\xi}^{(0)} = q$. Write $\Pi[\cdot]$ for projection back to the sphere, $b \in \mathbb{R}^N$ for a per-memory strength bias, $\beta > 0$ for an inverse temperature, and $\mathrm{lse}(\cdot)$ for the log-sum-exp. The modern Hopfield energy (Ramsauer et al., 2021) is

$$E_M(\boldsymbol{\xi}) \;=\; -\tfrac{1}{\beta}\,\mathrm{lse}\big(\beta(X^\top \boldsymbol{\xi} + b)\big) \;+\; \tfrac{1}{2}\|\boldsymbol{\xi}\|^2, \tag{1}$$

minimised by the concave–convex (CCCP) update, which we run for T steps:

$$\boldsymbol{\xi}^{(t+1)} \;=\; \Pi\Big[(1-\eta)\, X\ \mathrm{softmax}\big(\beta(X^\top \boldsymbol{\xi}^{(t)} + b)\big) \;+\; \eta\, R\big(\boldsymbol{\xi}^{(t)}\big)\Big]. \tag{2}$$

Equation (2) decreases (1) monotonically and converges for $\eta = 0$. The operator R is a *routing* map, discussed in §3.3.

3.2 Dense retrieval is a corner of this family

Proposition 1 (The reduction)**.** *Let the patterns z_i and the cue q be L_2-normalised, and set $T = 1$, $b = 0$, $\eta = 0$. Then as $\beta \to \infty$ the state after one update converges to the stored pattern of maximal cosine similarity to the cue,*

$$\lim_{\beta\to\infty} \Pi\big[X\ \mathrm{softmax}(\beta X^\top q)\big] \;=\; z_{i^\star}, \qquad i^\star = \arg\max_i \langle z_i, q\rangle,$$

whenever the maximiser is unique. The reduction is therefore exact *at $k = 1$. For $k > 1$ the dense top-k neighbourhood is recovered by reading the ranking off the* cue*, $\langle z_i, q\rangle$ — the standard definition, and the one our dense baseline uses — and* not *off the converged state: ranks $2..k$ of $\langle z_i, \boldsymbol{\xi}^{(1)}\rangle$ order the patterns by proximity to $z_{i^\star}$, not to q, because the state has forgotten the cue by construction. Remark 1 makes this precise, and §5.1 shows it is the mechanism behind the shortlist regression rather than a technicality.*

Proof sketch. With $b = 0$ the logits are $s_i = \langle z_i, q\rangle$. Let $\delta = s_{i^\star} - \max_{i \neq i^\star} s_i > 0$. Then $\mathrm{softmax}(\beta s)_{i^\star} = \big(1 + \sum_{i \neq i^\star} e^{-\beta(s_{i^\star} - s_i)}\big)^{-1} \geq (1 + (N-1)e^{-\beta\delta})^{-1} \to 1$, and every other coordinate is $O(e^{-\beta\delta})$. Hence $X\,\mathrm{softmax}(\beta X^\top q) \to z_{i^\star}$, and normalisation fixes the point since $\|z_{i^\star}\| = 1$. The full argument — with the explicit $O\big((N-1)e^{-\beta\delta}\big)$ rate, the tied-maximiser case, and the analysis of what happens for $k > 1$ — is in Appendix A. □

We treat Proposition 1 as an executable claim rather than a remark. The test suite instantiates the configuration above and asserts *top-1* agreement with an independent exact-cosine implementation on every query

— the quantity the proposition states, no more; it passes 1000/1000 on synthetic vectors (d=64) and 838/838 on real sentence embeddings (d=768), to floating-point precision, and it gates continuous integration. If it ever fails, no downstream number in this paper is trustworthy (X1).

Two consequences organise everything that follows.

First, the baseline cannot be disadvantaged. "Dense top-k" in every table below is Equation (2) with the arguments of Proposition 1, run through the same code path, the same encoder and the same candidate set as the method under test. The most common rejection reason for papers of this shape—that the authors tuned their own method and not the baseline—is unavailable here by construction.

Second, the research question becomes finite. The reduction identifies a specific point in a parameter space; the scientific content is which coordinates are worth moving.

3.3 Three axes away from the corner

Exactly three coordinates of Equation (2) are free, and each encodes a distinct hypothesis about what agent memory needs.

G1 — depth, $T > 1$. Iterating repairs a degraded cue. Cosine cannot imitate this: it has no mechanism for revisiting a cue.

G2 — basin depth, $b \neq 0$. A memory used constantly earns a deeper basin and therefore a wider one, so a vaguer cue still finds it. This folds usage, recency and surprise into a single term of the energy instead of three heuristics blended by hand outside it. Cosine can partially imitate it with an additive score bonus.

G3 — routing, $R \neq I$. Memories that co-occur in time pull on one another, so falling into one drags the state toward its companions. This is the only axis that can express a relation cosine structurally cannot represent, and it is the axis on which our one positive result lives. Critically, R must be fitted from temporal co-occurrence and never from embedding similarity: fitting it on similarity collapses G3 back into the baseline and silently invalidates the claim.

3.4 The β leverage band

Proposition 1 has a corollary that is easy to state and, we will argue, widely consequential.

Corollary 1 (High-β degeneracy)**.** *For any T, as $\beta \to \infty$ with $b = 0, \eta = 0$ the trajectory of Equation (2) is constant after the first step, because stored patterns are exact fixed points. Multi-step descent at high β is therefore* identical *to one-step descent, which by Proposition 1 is cosine retrieval.*

Empirically the degeneracy arrives early, and accuracy is the wrong way to see it. Two retrievers can score identically while disagreeing about which memory they return, so we measure *top-1 decision agreement* with cosine — the fraction of queries on which descent returns the exact memory cosine would have (Figure 1, `sweep_beta_transition.py`).

The curve has two failures and one working region. Below $\beta \approx 24$ the softmax is flat enough that the first step averages a large fraction of the bank and the state lands at its centroid: Recall@1 is 0.100 at β=12, which is exactly chance for a ten-class bank, and decision agreement with cosine is 1%. Above the band, descent converges on cosine's own answer — agreement rises through 86% at β=64 and 96% at β=128 to **99.4**% at β=512, where accuracy is 0.4986 against cosine's 0.5000. Between them is a genuine interior optimum: β=48 reaches 0.514, +0.014 over cosine, while still disagreeing with it on 27% of queries. The band is narrow — roughly $[32, 64]$ on this encoder, with useful leverage over perhaps a factor of two in β.

Three implications. (i) The apparent *plateau* in accuracy above $\beta \approx 64$, which a coarse sweep reads as a robust operating region, is the method dissolving into the baseline rather than working; the agreement curve distinguishes the two and the accuracy curve cannot. (ii) The band is a property of the encoder and must be re-measured per encoder: the interval above is for ResNet-18 features (d=512), while on `all-mpnet-base-v2`

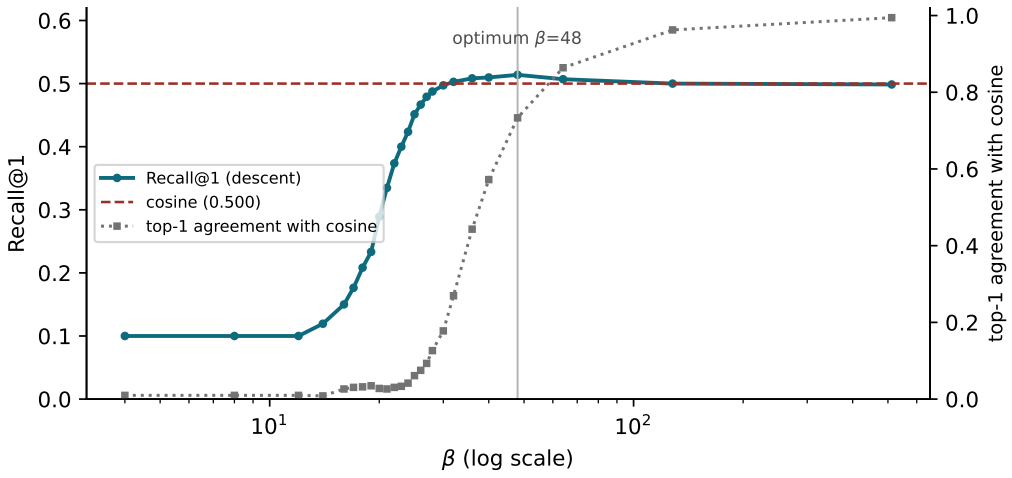


Figure 1: The β leverage band, on held-out image completion. **Left axis:** Recall@1 against the cosine baseline (dashed). **Right axis:** the fraction of queries on which descent returns the *same memory* as cosine. Below the band the state collapses to the corpus centroid and accuracy falls to chance ($0.100 = 1/10$); above it descent makes cosine's decision on 99.4% of queries. The usable interval is the narrow region where the method is neither collapsed nor dissolved.

(d=768) the lower edge sits higher — at β=16 the effective number of patterns in the blend is 377 (X11), against a threshold of ≈ 16 on isotropic synthetic vectors (X3). The threshold survives as a mechanism and dies as a number. (iii) Any experiment comparing "Hopfield retrieval" against "cosine retrieval" at high β is comparing cosine against cosine on nearly every query, and will correctly report no difference for the wrong reason. We flag this as a live hazard for this literature; we committed it ourselves, and §7 reports what it cost us.

3.5 Beyond the landscape: memory as a model of what follows what

Every mechanism in §3.3 except G3 changes how an answer is *read off* a fixed landscape. None of them changes the landscape. The results in §5.4 make that distinction empirically load-bearing: readout mechanisms uniformly fail on agent text.

G3 is different in kind, and the formulation we adopt for it is the joint-embedding predictive one (LeCun, 2022; Assran et al., 2023): predict the *representation* of the next episode, not its tokens. An agent's episode stream is a trajectory, and temporal adjacency is free supervision—no labelling is required to know which episode followed which.

Let $\mathcal{T}$ be the set of observed consecutive index pairs, Z_c the matrix of cue representations over $\mathcal{T}$, and $\Delta_i = z_{\text{next}(i)} - z_{\text{cue}(i)}$ the displacement each transition took. We consider two instantiations.

Parametric. A low-rank predictor trained with an InfoNCE objective against a frozen target encoder. This is the natural first choice and it fails completely (§5.3): it memorises the storylines it was fitted on.

Nonparametric. Carry the transitions rather than compressing them, and transfer displacements locally:

$$P(q) \;=\; \Pi\Big[q \;+\; \alpha \textstyle\sum_{i \in \mathcal{N}_k(q)} w_i \, \Delta_i\Big], \qquad w = \operatorname{softmax}\big(\tau \, Z_c^\top q\big)\big|_{\mathcal{N}_k(q)}, \tag{3}$$

where $\mathcal{N}_k(q)$ are the k nearest remembered cues and α is a step size. This is still a world model—it predicts what comes next in embedding space—and it composes: iterating it rolls the model forward. It is the one mechanism in this paper that beats cosine on held-out data.

The step size α is not cosmetic, and §7 explains why we are emphatic about it. A single remembered displacement has norm 1.157 against a unit-norm cue, so at α=1 the prediction is dominated by the displacement

and lands essentially on $z_{\text{next}(i)}$—a *stored training episode*, 100% of the time on our corpus against a bank prior of 14%. The kernel in (3) implicitly applies $\alpha \approx 0.34$ through cancellation among near-orthogonal displacements, and that shrinkage, not the weighting, is what keeps the cue in control of where the prediction lands (X23B). Any comparison between predictors that does not match effective step size measures step size.

4 Experimental Protocol

Protocol decisions are listed here in full because several of them changed a conclusion's sign, and §7 reports by how much.

Matched encoder. Every method under comparison—ours, every baseline, every ablation—is handed identical vectors from the same frozen model. Text uses `all-mpnet-base-v2` (Reimers & Gurevych, 2019), d=768; images use ImageNet-pretrained ResNet-18 features (He et al., 2016), d=512. Encoders are never fine-tuned, and no result in this paper can be attributed to representation quality.

Corpora.

- **Images (completion regime).** 400 Imagenette images as the memory bank; queries are occluded held-out images at 30/50/70% occlusion. Evaluation uses 5,847 query instances drawn from 1,949 distinct images, the full validation split.
- **LoCoMo (semantic-match regime).** 1,977 real questions over 5,882 conversational turns (Maharana et al., 2024). Metric: Coverage@5 over gold evidence turns.
- **LoCoMo (composition regime).** The same 5,882 turns, re-used as temporal-association probes: cue a turn, retrieve the turn that followed it in the same session. 5,233 probes at one hop and 5,013 at two, over 272 sessions. The candidate set is the full pooled bank of all ten conversations, with dialogue identifiers namespaced by conversation (Appendix E). This is the external-validity evidence for §5.3, and it is a corpus we did not build.
- **Episodic corpus (composition regime).** A purpose-built store of 1,319 episodes across 122 storylines with 146 multi-target questions, plus 900 compositional filler episodes and planted near-miss distractors. Association probes are generated from it as consecutive (or h-distant) within-storyline pairs: 257 probes at one hop and 135 at two. This corpus is ours, it is generated rather than collected, and on its own it would not support a claim; every composition result measured on it is reported beside the LoCoMo measurement of the same quantity.

Baselines. Dense top-k (Proposition 1), BM25 (Robertson & Zaragoza, 2009), hybrid reciprocal-rank fusion (Cormack et al., 2009), and—in the composition regime—a parametric JEPA predictor. All lexical and hybrid baselines receive the same tuning budget as our methods, stated per experiment in Appendix B.

Splits and selection. Hyperparameters are never selected on the data a number is reported on. In the composition regime the split unit is the *storyline* on our corpus and the *session* on LoCoMo, never the question: a model fitted on transitions from one storyline or session is never evaluated on it. Because the corpus holds only 122 storylines, a single 50/25/25 split leaves 61 one-hop and 30 two-hop test probes, intervals on which span most of the plausible range before the experiment begins. We therefore use thread-level 4-fold rotation: every storyline is in the test set exactly once and is never in the train or validation fold that tests it. Both the world model's own (k, τ) and the readout's parameters are selected per fold on that fold's validation storylines.

Intervals. Headline numbers are *paired* differences against the baseline on the same queries, with 95% bootstrap intervals (20,000 resamples) over the *cluster*—questions, storylines, or source images as appropriate—never over individual measurements. Comparing two overlapping marginal intervals is the wrong test when

Table 1: Image occlusion recovery, Recall@1. β selected on a disjoint tune half; one pre-committed paired test on the held-out half. n=5,847 query instances from 1,949 distinct images; bootstrap resamples images. The tune-half estimate was +0.016 against a held-out +0.014, indicating no selection bias.

Occlusion	**Cosine**	**Descent** (β=40)	**Paired Δ (95% CI)**
30%	0.730	0.733	+0.003 [-0.008, +0.014]
50%	0.542	0.559	+0.017 [+0.003, +0.031]*
70%	0.386	0.407	+0.021 [+0.007, +0.034]*
pooled	**0.553**	**0.566**	**+0.014** [+0.006, +0.021]*

both methods see the same queries; resampling rows rather than clusters is pseudo-replication and we quantify what it cost us in §7. Image-modality intervals resample *images*, not query instances, since instances from one image are correlated across occlusion levels and seeds.

Dimensionless hyperparameters. Every scale that crosses dimensionalities is normalised before use: a perturbation of nominal magnitude σ is applied as $\sigma/\sqrt{d}$, so that its displacement norm is dimension-independent. At d=768 the unnormalised form is wrong by a factor of 28.

Symmetric candidate sets. Where a cue is itself a stored memory—as in every association probe—it is excluded from the candidate set for *every* method identically, before ranking. Asymmetry here produces exact zeros that read as findings.

Latency. No accuracy number appears without its cost. Iterative descent at T=8 costs 21× cosine in measured wall-clock (0.0502 ms vs. 0.0024 ms per query), and multi-β ensembling a further 1.6–2.9×. We report both wall-clock and matmul counts, because they diverge: the integrated mixture issues the same 16 matmuls as single-β descent and costs 1.6× the wall-clock, since mixing m softmaxes per step is arithmetic outside the matmuls. Figures in Table 6 and Appendix B.

Record. Every experiment is logged in an append-only ledger with an identifier, the question it asked, the result, and what it changed. Entries are never edited after their result is in; an overturned entry becomes a new entry pointing back at it. Identifiers of the form Xn in this paper index that ledger, including the entries that overturn our own earlier claims.

5 Results

All numbers below are paired against the dense baseline of Proposition 1 on identical queries, with cluster-bootstrap intervals. Ledger identifiers index `docs/10_EXPERIMENT_LEDGER.md`.

5.1 Completion: energy descent beats cosine, by more as the cue degrades

Table 1 is the paper's one clean positive for the energy mechanism itself. The effect is small in absolute terms and grows with corruption severity exactly as Claim 1 predicts, with no effect where there is nothing to reconstruct. It costs 21× cosine's wall-clock latency (0.0502 ms vs. 0.0024 ms per query).

Two qualifications belong next to it rather than in a limitations section. First, the advantage is *capacity-dependent*: growing the bank from 100 to 4,000 patterns at fixed β=32, cosine overtakes descent past $N \approx 800$. Whether this is a β-scaling artefact or a genuine capacity limit is unresolved, and it bears directly on any claim about agent-scale memory. Second, descent wins at rank 1 and *loses badly on the shortlist*: Recall@5 is 0.605 against cosine's 0.759 and MRR 0.589 against 0.643 (Table 6). Converging to a single attractor means ranks 2–5 measure proximity to that attractor rather than to the query. A better top-1 and a worse shortlist is a different product decision from a uniformly better retriever, and reporting one without the other would misrepresent the method.

Table 2: LoCoMo, 1,977 questions over 5,882 turns. `anchor` interpolates between pure cosine (1.0) and pure descent (0.0); the baseline is a point in our own parameter space, so the sweep measures exactly what descent adds. The reduction check passes 196/196 at `anchor`=1.0.

Condition	Coverage@5	95% CI
`anchor`=1.00 ***(is cosine)***	**0.336**	[0.316, 0.355]
`anchor`=0.90, pp_0=32	0.331	[0.311, 0.351]
`anchor`=0.75, pp_0=4	0.321	[0.301, 0.340]
`anchor`=0.50, pp_0=4	0.304	[0.285, 0.324]
`anchor`=0.25, pp_0=4	0.273	[0.255, 0.292]
`anchor`=0.00, pp_0=4	0.252	[0.234, 0.271]
energy, fixed β=32	0.236	[0.218, 0.254]

An earlier verdict of ours, withdrawn. A previous version of this study used 80 distinct images with unpaired intervals roughly 5× the effect size and β tuned on the test images. Its conclusions were underpowered and are withdrawn; Table 1 supersedes them.

5.2 Semantic match: every increment of descent costs accuracy

Table 2 is monotone: the optimum sits at the boundary where the method *is* the baseline. The table shows the best pp_0 at each anchor; the worst cell of the full sweep is `anchor`=0.00, pp_0=32 at 0.201 [0.184, 0.218], which is the broad-start annealing failure of §3.4 compounding with the absent query anchor. There is no interior operating point at which descent helps on question→evidence retrieval, which is the dominant query type in deployed agent memory.

A data bug that invalidated an earlier result. LoCoMo dialogue identifiers are unique only *within* a conversation: 5,882 turns carry 1,033 distinct identifiers, and 132 of conversation 0's 134 gold identifiers recur in other conversations. Our first capacity sweep pooled banks without namespacing, so distractors satisfied gold identifiers and scored as hits. Its headline—"adaptive β improves as the bank grows"—was an artefact; with namespaced identifiers adaptive β *degrades* (0.250 → 0.231), merely less steeply than fixed β (0.244 → 0.216). We report this because the failure mode is specific to this widely used benchmark and will recur.

5.3 Composition: the one mechanism that works

The gate. Before building a predictor we checked whether cosine already solves "what happened next". It does not: Recall@1 is 0.105 with the gold episode at median rank 24 of 1,319. The failures are instructive—they are lexical collisions across unrelated storylines, where a cue about a cottage drive returns "forty minutes each way" from an apartment-hunting thread. Oracle-restricting candidates to the correct storyline still reaches only 0.488, so the difficulty is not merely retrieval breadth. Temporal succession is genuinely not a similarity relation (X16).

The parametric predictor fails completely. A low-rank JEPA predictor fitted on some storylines and evaluated on others reaches Recall@1 of 1.000 on the storylines it was fitted on and 0.000 on unseen ones, against the cosine baseline's 0.131—worse than doing nothing. No capacity escapes it: across ranks 2 to 32 the best unseen Recall@1 at any setting is 0.016, and the two largest capacities score exactly zero. The diagnostic explains why: displacement vectors between consecutive episodes are near-orthogonal (mean pairwise cosine +0.004) and the top ten principal components carry only 31% of their variance. There is no global "and then" direction to fit, and 134 transitions in 768 dimensions cannot teach one (X17, re-run as X17-R). We report this as a result about inductive bias, not as a tuning failure.

The nonparametric predictor works, on our corpus and on a public one. Equation (3), with (k, τ) selected on held-out storylines or sessions only, beats cosine on one-hop association in three independent

Table 3: One-hop temporal association. Three implementations, two corpora. "Clusters" is the bootstrap resampling unit—storylines for the episodic corpus, sessions for LoCoMo. Implementations (ii) and (iii) were written without reference to (i). Absolute scores are lower on LoCoMo because the task is harder: a larger pooled bank, nine conversations of distractors, and speaker-alternating turns. The relative effect is *larger* there.

	Corpus	Probes	Clust.	Cosine	WM	Paired Δ R@1
(i)	episodic (ours), 50/50	123	61	0.106	0.130	+0.024 [-0.032, +0.083], p=0.24
(ii)	episodic (ours), 4-fold	257	122	0.105	0.140	+0.035 [+0.012, +0.061], p=0.002
(iii)	**LoCoMo**, 4-fold	**5,233**	**272**	**0.064**	**0.107**	**+0.043** [+0.036, +0.050], $\mathbf{p < 10^{-4}}$

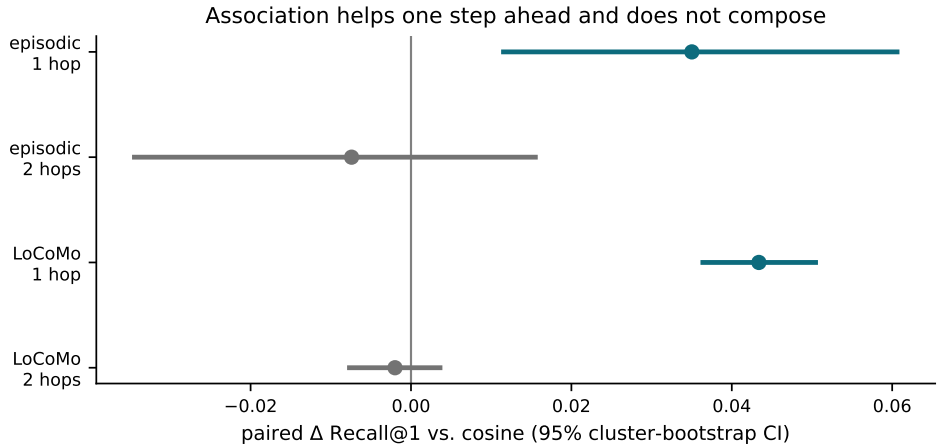


Figure 2: Paired Δ Recall@1 against cosine, both corpora, one and two hops. Intervals are 95% cluster bootstrap (storylines for the episodic corpus, sessions for LoCoMo). The one-hop effect is positive on both corpora and larger on the public, harder one; the two-hop effect is null on both. Generated by `scripts/make_tables.py`.

implementations (Table 3). The third is the one that matters: it runs the same probe design on LoCoMo, a benchmark built by another group, against the *full pooled bank* of 5,882 turns from all ten conversations, so a prediction must beat distractors from nine unrelated conversations as well as its own. Dialogue identifiers are namespaced by conversation—without that, 5,882 turns carry only 1,033 distinct identifiers and distractors satisfy gold identifiers, the bug described in §5.2.

On LoCoMo the effect is +67% relative at Recall@1 and +0.077 [+0.067, +0.087] at Recall@5 (0.174 → 0.251). This is the only claim in the paper with external validity, and the only one reproduced three times; we weight it accordingly, and we have restricted the paper's positive claims to it.

Multi-hop composition does not hold. We withdraw our own earlier result. An earlier run of ours indicated that iterating the predictor recovers two-hop chains it was never trained on: Recall@5 of 0.446 against cosine's 0.262, $\Delta = +0.185$ [+0.076, +0.297], $p < 0.001$. That was the strongest result we had recorded. It does not survive.

Three subsequent measurements (Table 4) return nulls, the last of them on public data at n=5,013 with 272 clusters—roughly 37× the probe count of the original. We therefore withdraw the two-hop claim as an artifact rather than suspending it. The +0.185 figure should not be cited, including from our own ledger, where it now carries a pointer to this result.

A residual signal does exist at two hops on LoCoMo at Recall@5 (+0.012 [+0.004, +0.019]), an order of magnitude below the one-hop effect and not visible at Recall@1. We report it for completeness and do not

Table 4: Two-hop composition, every measurement we have. The original script was lost before it was committed, so (a) cannot be re-run; (b) and (c) are independent reimplementations under storyline- and session-disjoint rotation. Row (c) is p=0.77; the claim is withdrawn.

	Corpus / protocol	Probes	Metric	Result
(a)	episodic, 50/50 *(lost)*	65	R@5	0.446 vs. 0.262; Δ+0.185
(b)	episodic, 4-fold	135	R@5	0.230 vs. 0.207; spans zero
(c)	LoCoMo, 4-fold	5,013	R@1	0.041 vs. 0.043; $\Delta - 0.002$ [-0.008, +0.004]

build on it. The honest summary is that association helps one step ahead and does not compose: the error introduced by predicting from a prediction exceeds whatever compositional signal the second step carries.

Why we report this at length. The withdrawn result was, for several weeks, the headline of this project and the organising claim of a design document built on top of it. It was overturned by reimplementation, not by review—twice—and the version that finally settled it is the public-benchmark run that a reviewer would have demanded anyway. We regard the sequence as the strongest available evidence for §7's argument that independent reimplementation is a distinct and stronger property than peer inspection.

5.4 Eight mechanisms that do not pay for themselves

Two findings survive from the controls in Table 6. The integrated $[1, 5, 20]$ control collapsed to exact chance ($0.108 = 1/10$) with *identical* accuracy at every occlusion level: one bad scale inside a shared trajectory poisons the entire state. The ensemble version of the same scales reaches 0.282, because there a bad scale only dilutes one vote among independent runs. This is direct evidence that the coupling between scales is real and can hurt as easily as it can fail to help.

5.5 The multi-modality result, and why it does not help

The predictor of Equation (3) returns a weighted average. A natural hypothesis, and one we committed to in a design document before testing it, is that this is a mistake: if a cue has several legitimate successors, their average is a point in embedding space where nothing is stored—a ridge between basins rather than a basin. Preserving the modes should therefore help, and should motivate a stochastic (Boltzmann) rather than deterministic relaxation (Ackley et al., 1985; Welling & Teh, 2011).

The premise is correct. Among the top three modes of $P(\cdot \mid q)$ the mean mutual cosine is 0.353, and 97% of cues have modes that mutually disagree (X24). There is genuinely a distribution for averaging to destroy.

The conclusion is wrong. With step size matched (§3), the mixture readout

$$\text{score}(m) \;=\; \text{lse}_s\big[\beta\langle \boldsymbol{\xi}_s, z_m\rangle + \log w_s\big]$$

loses to the averaged predictor at one hop, $\Delta = -0.062$ [-0.094, -0.034], and ties at two (X25). The off-manifold gap the hypothesis rested on is also much smaller than assumed: mean nearest-neighbour cosine is 0.482 for a real stored episode against 0.462 for an averaged prediction, a gap of -0.021.

The mechanism is the step size. Averaging does two things at once, and the one that matters is not the one the hypothesis named: it shrinks the displacement by a factor of 0.34 through cancellation among near-orthogonal transitions, and that shrinkage keeps the cue in control of where the prediction lands. Preserving the modes removes the shrinkage. On this data, averaging is not a failure to be fixed—it is what makes the predictor work.

We report this at length because it is the clearest instance in the paper of a correct premise licensing a wrong design, and because the first version of the experiment was confounded by our own implementation in exactly the way §7 describes.

Table 5: Negative results, each with its diagnosed mechanism. "Regime" indicates where the test was run. Two of these overturn claims we previously made ourselves.

Mechanism	Result	Diagnosed mechanism
Multi-step descent (G1)	no effect	Identical to one step above the β threshold—precisely what Corollary 1 predicts. Once β commits on step one there is nothing left to do (X2).
Residual peeling	0.268 vs. repulsion 0.290	Restores multi-answer recall on orthogonal synthetic patterns (0.258 → 1.000) and inverts on real embeddings: non-orthogonal golds mean projecting one out damages the others (X5, X13).
Repulsive parallel descent	+0.004, p=0.43	Survives to 146 questions but not to held-out selection. The +0.042 trend was hyperparameter overfitting (X15).
Noise-seeded parallel descent	fails at every σ	No intermediate value exists. Too little noise and all chains land in one basin; too much and they scatter uniformly, where correct answers are no likelier than any others. Isotropic noise carries no information about where the other answers are (X7).
Cosine-seeded parallel descent	exactly chance	Circular. Stored patterns are exact fixed points, so descent is a no-op; ranking the landing sites requires a signal the descent has already destroyed. Ranking them by similarity to the cue scores 1.000—but that ranking *is* cosine top-k (X7).
Multi-scale β	+0.0003 [-0.0014, +0.0021]	Expressively distinct and performance-neutral. An integrated mixture is provably not reducible to any single β (Yang et al., 2018), and diverges from an ensemble at T=8 (63% top-1 agreement), yet changes *which* queries are answered, not how many (`exp_multibeta_scales.py`; this study predates the ledger and has no X identifier).
Per-memory / learned β	inert	The learned values (51–88) land above the leverage band, and a shuffle control with the same values randomly reassigned scores identically. Conditioning β on the query is motivated—the optimum does differ by corruption level—but the learned function converged to a near constant (β≈31 everywhere).
Predictor coupled into the energy	never exceeds external	Interpolating the routing operator inside the descent step gives 0.130 at η=0.5 and 0.154 at η=1.0—the latter merely *recovering* the external predictor, since η=1 replaces the Hopfield step entirely (X18). This was our stated novelty against prior work (Dury, 2026) and it is withdrawn.

6 The Regime Taxonomy

The results in §5 flip sign between modalities. We claim this is not a modality effect but a *query-type* effect, and that one sentence predicts the sign in advance.

Claim 1 (The completion premise)**.** Associative descent improves retrieval when, and only when, the cue is a degraded instance of a stored pattern. The descent step computes a convex combination of stored patterns and moves the state onto their span. Where the cue is a corrupted copy of a memory, that motion restores signal the corruption removed. Where it is not, the same motion destroys the cue's discriminative content with nothing to recover.

Claim 1 is falsifiable in two directions and we test both.

Table 6: Multi-β scales on image completion, with compute, ordered by Recall@1. Bootstrap resamples images. Controls were pre-registered to collapse and did. Note the R@5/MRR columns: descent's shortlist is substantially worse than cosine's at every setting, which is the shortlist regression of §5.1 and Remark 1. Single β beats cosine by $+0.0130$ $[+0.0074, +0.0186]$; no mixture beats single β.

Condition	**R@1**	**R@5**	**MRR**	**matmuls**	**Δ vs. single β**
integrated mixture [32,40,48]	0.559	0.591	0.584	16	+0.0036 [-0.0033, +0.0106]
ensemble [32,40,48]	0.559	0.599	0.586	48	+0.0032 [-0.0038, +0.0103]
single beta=56	0.555	0.605	0.589	16	—
cosine	0.542	0.759	0.643	1	—
control ensemble [1,5,20]	0.282	0.296	0.303	—	collapse (predicted)
control integrated [1,5,20]	0.108	0.318	0.217	—	collapse (predicted)

Table 7: The three regimes, the operating point each requires, and the evidence. No single fixed configuration serves more than one.

Regime	**The cue is**	**What wins**	**Operating point**
Completion	a degraded copy of a stored pattern	multi-step descent	$T > 1$, β in band, H=0
Composition	h steps upstream of the answer	predictor rollout	$H \geq 1$
Semantic match	a question about stored evidence	plain cosine	T=1, $\beta \to \infty$

Satisfied, and descent wins, more as the premise is satisfied harder. An occluded image *is* a degraded copy of a stored image. Descent's advantage is null at light corruption, where little has been removed ($+0.003$, $[-0.008, +0.014]$, n.s. at 30%), and grows monotonically with severity to $+0.021$ $[+0.007, +0.034]$ at 70%. A dose-response relationship in the predicted direction, on a pre-committed held-out split.

Violated, and descent loses, more as descent is applied harder. A LoCoMo question is not a corrupted conversational turn. Interpolating from pure cosine to pure descent costs accuracy at every increment, $0.336 \to 0.252$, with the optimum at the cosine boundary. A dose-response relationship in the predicted direction again, with the reduction check passing 196/196 at the boundary so that the monotonicity is a property of the method rather than an implementation artefact.

6.1 Three regimes

Claim 1 partitions agent retrieval into regimes that want mutually exclusive operating points (Table 7).

The practical consequence is sharper than a taxonomy usually is. A deployed agent meets all three regimes in one workload, so any system with a *fixed* retrieval configuration must lose two of them. This is an argument for query-conditional inference—selecting the operating point per query rather than per system—which we state as the natural next architecture. We have since built and evaluated the narrowest version of it, conditioning β alone, and **it does not work.** Setting β per query from the inverse top-two margin or the inverse score dispersion beats a fixed β=16 by $+0.034$ $[+0.004, +0.066]$ and $+0.067$ $[+0.038, +0.097]$—and loses to the best fixed β, chosen on training folds only, by -0.121 [-0.149, -0.098] and -0.089 [-0.121, -0.061] (X65-R). The apparent gain was the reference point: β=16 scores 0.241 where the fold-selected β=32 scores 0.397, so conditioning the knob recovers part of a handicap rather than adding anything. A sign-reversed control ($\beta \propto$ margin) costs -0.390 [-0.433, -0.348], so the rules do read the signal they were designed to read; they simply cannot spend it.

That is a result about one knob, not about the taxonomy, which asks for selection among *regimes*—and that remains unbuilt. But it is a pointed one, because this is the component we flagged as most exposed to the selection error of §7: "choose the budget that maximises the score" is test-set tuning with extra steps unless the selector is fitted and frozen on training data. What went wrong first was not the selector but the *baseline* it was scored against.

6.2 What the taxonomy predicts that we have not yet run

The cheapest discriminating test is a text occlusion study: hold the LoCoMo corpus and encoder fixed and swap the queries from questions to *masked turns*. Only the query type changes. Claim 1 predicts that descent regains its advantage; if it does not, the premise is wrong and the modality explanation is right. We flag this as the experiment most likely to falsify our own central claim, and its absence as a limitation rather than as future work.

7 What the Protocol Was Worth

This section reports six errors we made, and for each one the magnitude of the false result it produced in our own data. We include it because the errors are not exotic—each is a default that a reasonable person reaches for—and because their cost here is large enough to flip conclusions. The point is not confession. It is that for this class of experiment, the protocol is not a hygiene matter around the result; it *is* the result, and we can put numbers on that.

Three observations we think generalise beyond this paper.

The errors are not symmetric in how they feel. Two of the six produced false *negatives*, and those were by far the harder to find, because a null result invites no scrutiny. The dimensionless-hyperparameter bug reported a working mechanism as 0.016 when its true value was 0.277; it was found only because the number was suspiciously round. A field that publishes negative results—which we are arguing it should—inherits an obligation to audit them as hard as positives.

Self-replication caught more than review would have. The two-hop result in §5.3 and the mode-averaging result in §5.5 were both contradicted by reimplementation, not by inspection. In both cases the original code had been read, reviewed and believed. We now treat "has this been written twice, independently" as a distinct and stronger property than "has this been reviewed", and we report which of our claims have it: one does.

An uncommitted script is not a result. The sixth error is different in kind from the other five: it cannot be corrected, only recorded. Both drivers that were lost have now been rewritten, and rewriting them is what revealed that one of the two results did not reproduce. We state the rule the episode produces, because it is cheap and we did not follow it: a number does not enter the record until the code that produced it is committed.

The reduction is what made the audit possible. Because the baseline is a configuration of the same code (Proposition 1), four of the six errors above were detectable as a *violation of a known equivalence*—the method failing to reproduce cosine where it provably must. An architecture in which every component has a null setting that reduces to the component below it converts a class of silent bugs into failing assertions. We regard this as the main transferable engineering claim of the paper, separate from any empirical finding: build the baseline as a corner of the method, and assert the reduction in continuous integration.

8 Limitations

One public benchmark, one task shape. The one-hop positive replicates on LoCoMo (Maharana et al., 2024), which discharges the external-validity concern for that result. It does not discharge it generally. The probe shape—cue a turn, retrieve the turn that followed—is one operationalisation of temporal association, and LoCoMo's alternating two-speaker structure gives next-turn prediction a regularity that a single agent's episode stream may not share. LongMemEval (Wu et al., 2025) is the obvious second corpus and we have not run it.

Association does not compose, and we do not know why. Two hops is null on both corpora (§5.3). We attribute it to compounding prediction error exceeding the compositional signal, but that is an inter-

Table 8: Six protocol errors and what each was worth. "Reported" is the number the error produced; "corrected" is the number after the fix. Every one would have been publishable.

Error	Reported	Corrected	Direction and why
Hyperparameter not dimensionless	0.016	0.277	**False negative.** At d=768, a nominal σ=0.35 perturbation applied without a $1/\sqrt{d}$ factor has norm 9.7 against a unit cue, giving $\cos(q, q+\epsilon)$=0.124: the cue is destroyed. The method appeared not to work.
Pseudo-replicated bootstrap	p=0.0005	p=0.076	**False positive.** 185 rows were 5 correlated seeds over the same 37 questions. Resampling rows treats them as independent; resampling questions is the correct cluster. A 150× difference in p.
Test-set hyperparameter selection	+0.042	+0.004 (p=0.43)	**False positive.** β was swept on the questions it was then scored on. Two corrections—more questions, then a storyline-disjoint split—took the effect to zero in stages (+0.042 → +0.027 → +0.004).
Asymmetric candidate exclusion	0.000 exactly	—	**False negative, disguised as a finding.** The cue was excluded from the candidate set for the baseline but not for the method under test. An exact zero alongside an unchanged Recall@5 looked like a structural result.
Unmatched step size	−0.115	−0.062	**Inflated effect.** Comparing predictor variants at α=1 against an average that implicitly applies $\alpha\approx$0.34 confounds the comparison with step size. Roughly half the apparent effect was the confound.
Uncommitted script	+0.049 (p=0.032)	+0.024 (p=0.24)	**Unresolvable.** A significant result at n=37 whose driver was never committed. Re-running the same split recovers the cosine baseline exactly but yields n=123 probes and no significant effect. With the original code gone the discrepancy cannot be diagnosed — only reported.

pretation rather than a measurement: we have not isolated the error accumulated per step, and the small residual Recall@5 effect on LoCoMo is unexplained. A rollout-depth sweep with per-step error attribution is the experiment this calls for.

The retrieval task is not the downstream task. Every number here is retrieval quality. Whether a +0.043 Recall@1 improvement on next-turn association changes what an agent actually does—answer quality, task success—is untested, and the mapping from retrieval metrics to agent behaviour is weak enough that we make no claim about it.

Scale. Banks run from 400 to 5,882 patterns with encoders of 512 and 768 dimensions. The image advantage is explicitly capacity-dependent—cosine overtakes descent past $N \approx 800$ at fixed β—so nothing here licenses a claim at the 10^6-memory scale a long-lived agent would reach. Whether the crossover is a β-scaling artefact is unresolved and we flag it as the most important unanswered question about the completion result.

The latency cost is real, and larger than the matmul count suggests. Descent at T=8 issues 16 matmuls against cosine's 1, but measured wall-clock is 21× (0.0502 ms vs. 0.0024 ms per query), for a pooled +0.014 Recall@1 in the one regime where it wins. Multi-β variants cost a further 1.6–2.9× for an effect indistinguishable from zero. We do not think this trade is attractive in deployment and say so.

The shortlist regression. Descent improves rank 1 and degrades ranks 2–5 (0.605 vs. 0.759 Recall@5). For an agent that pastes five memories into context, which is the standard usage, this is plausibly a net loss. Our positive result is therefore narrower than "better retrieval" even within the completion regime.

Biological framing is analogy. We use the words basin, attractor and consolidation because they are the established vocabulary of the formalism. We make no claim of correspondence with neural mechanism, and no result here depends on one.

The remaining axes have now been tested, and none of them helps. An earlier version of this section listed G2 and the controller as unevaluated. Both have since been measured, on a multi-target variant of the same LoCoMo corpus—retrieve the full set of annotated evidence turns for each of 1,977 questions—with the conversation as the resampling cluster. Ten clusters floor a blocked-permutation p at $2/2^{10} = 0.002$, so we give intervals and not p-values.

G2 (strength-modulated basin depth) hurts under every prior we tried—recency, centrality, session size, turn length—and hurts monotonically in the bias strength. The gentlest setting is indistinguishable from no bias at all (-0.004 [-0.035, +0.030]) and the strongest costs -0.495 [-0.533, -0.459] (X62). **G3 with a *fitted* routing operator** reproduces the X18 verdict by an independent route: diagonal routing costs -0.029 [-0.046, -0.013] at its mildest and dense routing -0.471 [-0.512, -0.424] at its strongest, so R=I—the corner—is the best routing we have been able to find (X63). The lifecycle operations built on G2, decay and consolidation, are still implemented and unevaluated; that part of the gap stands.

A single encoder per modality. Proposition 1 is encoder-independent, but the β band is not, and we measured it on one text encoder and one vision backbone. The claim that the band must be re-measured per encoder is supported by two points (synthetic and `all-mpnet-base-v2`), which establishes that it varies but not how it varies.

9 Conclusion

We set out to build a better memory for autonomous agents on associative-memory foundations. That thesis is refuted here by our own measurements, at adequate power, with the mechanism identified: every departure from the dense-retrieval corner that changes only how an answer is read off the energy landscape fails on agent text, because none of them changes the landscape.

What replaced it is narrower and better supported. Dense top-k retrieval is exactly the (T=1, $\beta \to \infty$, b=0, R=I) corner of the modern Hopfield update, which makes the standard baseline a configuration of the method under test rather than a rival outside it, and converts an ill-posed question into a finite one. Within that frame, the sign of the effect is predictable before the experiment: associative descent helps when the cue is a degraded instance of a stored pattern, by more as the degradation increases, and hurts monotonically when it is not. Of eight mechanisms beyond the corner, one survives—a nonparametric predictor over the agent's own observed transitions, which retrieves by succession rather than resemblance. It beats cosine on one-hop association in three independent implementations across two corpora, including +0.043 Recall@1 [+0.036, +0.050] on a public benchmark against a pooled bank of 5,882 turns. The same experiment bounds it: at two hops the effect is -0.002, and a +0.185 two-hop result of our own is withdrawn as an artifact. Association helps one step ahead and does not compose.

We also report what the protocol was worth, in the only currency that makes the argument: six errors, each a reasonable default, each worth a conclusion. Two produced false negatives, which were materially harder to detect than the false positives. Two of our own recorded claims were overturned by reimplementation

rather than by review—and the one that had been our headline was settled only by the public-benchmark run that external validity demanded anyway.

The constructive proposal we take from this is structural rather than empirical. Build the baseline as a corner of the method, give every added component a null setting that reproduces the configuration beneath it, and assert those equivalences in continuous integration. Four of the six errors above surfaced as violations of a known equivalence rather than as anomalous numbers. In a literature where the operator under test and the operator it is compared against coincide at the default operating point, that is not a convenience. It is the difference between measuring a method and measuring a bug.

Author Contributions

Pushkal Gupta led the project and is responsible for the substance of this paper. He formulated the reduction (Proposition 1) and its corollary, derived the proof in Appendix A, designed and implemented the energy core, the nonparametric predictor, the mixture readout, and the evaluation harness, and ran and analysed every experiment reported here, including the LoCoMo external-validity study and the protocol-error audit of §7. He built the experiment ledger and the generate-never-type table pipeline, and wrote the paper.

Rajit Agrawal maintained the bibliography and the citation audit, and contributed to the experimental-protocol documentation.

Anshveer Singh contributed the initial image-modality test harness and assisted with running the occlusion experiments.

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References

David H. Ackley, Geoffrey E. Hinton, and Terrence J. Sejnowski. A learning algorithm for boltzmann machines. *Cognitive Science*, 9(1):147–169, 1985. doi: 10.1016/S0364-0213(85)80012-4.

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15619–15629, 2023. doi: 10.1109/CVPR52729.2023.01499.

A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008. doi: 10.1162/rest.90.3.414.

Gordon V. Cormack, Charles L. A. Clarke, and Stefan Büttcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 758–759, 2009. doi: 10.1145/1571941.1572114.

Jason Dury. Predictive associative memory: Retrieval beyond similarity through temporal co-occurrence. *arXiv preprint arXiv:2602.11322*, 2026.

Kawin Ethayarajh. How contextual are contextualized word representations? comparing the geometry of bert, elmo, and gpt-2 embeddings. In *Conference on Empirical Methods in Natural Language Processing*, pp. 55–65, 2019. doi: 10.18653/v1/D19-1006.

Jun Gao, Di He, Xu Tan, Tao Qin, Liwei Wang, and Tie-Yan Liu. Representation degeneration problem in training natural language generation models. In *International Conference on Learning Representations*, 2019.

Taosha Guo and Fabio Pasqualetti. Learnable sequential memory in coupled oscillator networks. *arXiv preprint arXiv:2607.18439*, 2026.

Bernal Jiménez Gutiérrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. Hipporag: Neurobiologically inspired long-term memory for large language models. In *Advances in Neural Information Processing Systems*, pp. 59532–59569, 2024.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 770–778, 2016. doi: 10.1109/CVPR.2016.90.

John J. Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558, 1982. doi: 10.1073/pnas.79.8.2554.

Dmitry Krotov and John J. Hopfield. Dense associative memory for pattern recognition. In *Advances in Neural Information Processing Systems*, 2016.

Yann LeCun. A path towards autonomous machine intelligence. Position paper, Version 0.9.2, OpenReview, June 2022.

Chong Li, Taiping Zeng, Xiangyang Xue, and Jianfeng Feng. Dynamic manifold hopfield networks for context-dependent associative memory. *arXiv preprint arXiv:2506.01303*, 2025.

Qiang Liu and Dilin Wang. Stein variational gradient descent: A general purpose bayesian inference algorithm. In *Advances in Neural Information Processing Systems*, 2016.

Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of llm agents. In *Annual Meeting of the Association for Computational Linguistics*, pp. 13851–13870, 2024. doi: 10.18653/v1/2024.acl-long.747.

Stéphane G. Mallat and Zhifeng Zhang. Matching pursuits with time-frequency dictionaries. *IEEE Transactions on Signal Processing*, 41(12):3397–3415, 1993. doi: 10.1109/78.258082.

Beren Millidge, Tommaso Salvatori, Yuhang Song, Thomas Lukasiewicz, and Rafal Bogacz. Universal hopfield networks: A general framework for single-shot associative memory models. In *International Conference on Machine Learning*, 2022.

Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. Memgpt: Towards llms as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Thomas Adler, Lukas Gruber, Markus Holzleitner, Milena Pavlović, Geir Kjetil Sandve, Victor Greiff, David Kreil, Michael Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations*, 2021.

Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Conference on Empirical Methods in Natural Language Processing*, pp. 3980–3990, 2019. doi: 10.18653/v1/D19-1410.

Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: Bm25 and beyond. *Foundations and Trends in Information Retrieval*, 4(1–2):1–174, 2009. doi: 10.1561/1500000019.

Max Welling and Yee Whye Teh. Bayesian learning via stochastic gradient langevin dynamics. In Lise Getoor and Tobias Scheffer (eds.), *Proceedings of the 28th International Conference on Machine Learning (ICML-11)*, pp. 681–688, New York, NY, USA, 2011. ACM. ISBN 978-1-4503-0619-5.

Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. Longmemeval: Benchmarking chat assistants on long-term interactive memory. In *International Conference on Learning Representations*, 2025.

Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. A-mem: Agentic memory for llm agents. In *Advances in Neural Information Processing Systems*, pp. 20004–20031, 2025.

Zhilin Yang, Zihang Dai, Ruslan Salakhutdinov, and William W. Cohen. Breaking the softmax bottleneck: A high-rank rnn language model. In *International Conference on Learning Representations*, 2018.

Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. Memorybank: Enhancing large language models with long-term memory. In *AAAI Conference on Artificial Intelligence*, volume 38, pp. 19724–19731, 2024. doi: 10.1609/aaai.v38i17.29946.

Jinchang Zhu, Jindong Li, Cheng Zhang, Jiahong Liu, and Menglin Yang. Hela-mem: Hebbian learning and associative memory for llm agents. *Proceedings of ACL*, 2026.

A Proof of Proposition 1 and Corollary 1

Throughout, $X = [z_1 \cdots z_N] \in \mathbb{R}^{d\times N}$ with $\|z_i\| = 1$, the cue satisfies $\|q\| = 1$, and we set $b = 0$, $\eta = 0$, $T = 1$. Write $s_i = \langle z_i, q\rangle \in [-1, 1]$ for the logits, $p(\beta) = \mathrm{softmax}(\beta s) \in \Delta^{N-1}$, and $\boldsymbol{\xi}(\beta) = Xp(\beta)$ for the unnormalised state after one update.

A.1 The unique-maximiser case

Lemma 1 (Softmax concentration)**.** *Let $i^\star = \arg\max_i s_i$ be unique and let $\delta = s_{i^\star} - \max_{i\neq i^\star} s_i > 0$ be the top-two gap. Then*

$$1 - p_{i^\star}(\beta) \;\le\; (N-1)\, e^{-\beta\delta}, \qquad \textit{and} \qquad \sum_{i\neq i^\star} p_i(\beta) \;\le\; (N-1)\, e^{-\beta\delta}.$$

Proof. $p_{i^\star}(\beta)^{-1} = 1 + \sum_{i\neq i^\star} e^{-\beta(s_{i^\star} - s_i)} \le 1 + (N-1)e^{-\beta\delta}$, so $p_{i^\star} \ge (1 + (N-1)e^{-\beta\delta})^{-1} \ge 1 - (N-1)e^{-\beta\delta}$, using $(1+u)^{-1} \ge 1 - u$ for $u \ge 0$. The second bound is $1 - p_{i^\star}$. □

Lemma 2 (State convergence with rate)**.** *Under the hypotheses of Lemma 1, $\big\|\boldsymbol{\xi}(\beta) - z_{i^\star}\big\| \;\le\; 2(N-1)\, e^{-\beta\delta}$. Consequently $\boldsymbol{\xi}(\beta) \to z_{i^\star}$ as $\beta \to \infty$, and since $\|z_{i^\star}\| = 1$ the projection Π is continuous at the limit and $\Pi[\boldsymbol{\xi}(\beta)] \to z_{i^\star}$.*

Proof. Write $\varepsilon = \sum_{i\neq i^\star} p_i = 1 - p_{i^\star}$. Then $\boldsymbol{\xi} - z_{i^\star} = (p_{i^\star} - 1)z_{i^\star} + \sum_{i\neq i^\star} p_i z_i$, so by the triangle inequality and $\|z_i\| = 1$, $\|\boldsymbol{\xi} - z_{i^\star}\| \le \varepsilon + \varepsilon = 2\varepsilon$. Apply Lemma 1. For the last statement, $\|\boldsymbol{\xi}\| \ge \|z_{i^\star}\| - 2\varepsilon = 1 - 2\varepsilon > 0$ for β large, so Π is well defined and Lipschitz in a neighbourhood of $z_{i^\star}$. □

This proves the state half of Proposition 1, with an explicit $O(e^{-\beta\delta})$ rate governed by the top-two similarity gap. Note the rate depends on δ, not on N except logarithmically: $\beta \ge \delta^{-1}\log\big((N-1)/\epsilon\big)$ suffices for $\|\boldsymbol{\xi} - z_{i^\star}\| \le 2\epsilon$.

A.2 The $k > 1$ case: ranking, not state

Proposition 1 as used in this paper is a statement about the *top-k set*, which requires a separate argument because the state converges to a single pattern and the ranking is read off that state. Let π_β be the ordering of $\{1..N\}$ by $\langle z_i, \Pi[\boldsymbol{\xi}(\beta)]\rangle$ descending, and let $\pi_{\cos}$ be the ordering by $\langle z_i, q\rangle$ descending.

Lemma 3 (Rank-1 agreement)**.** *Under the hypotheses of Lemma 1, for β sufficiently large $\pi_\beta(1) = \pi_{\cos}(1) = i^\star$.*

Proof. By Lemma 2, $\langle z_{i^\star}, \Pi[\boldsymbol{\xi}]\rangle \to 1$ while for $j \neq i^\star$, $\langle z_j, \Pi[\boldsymbol{\xi}]\rangle \to \langle z_j, z_{i^\star}\rangle \le 1$ with equality only if $z_j = z_{i^\star}$, excluded by distinctness of the patterns. Choosing β past the point where the $O(e^{-\beta\delta})$ perturbation is smaller than half the gap $1 - \max_{j\neq i^\star}\langle z_j, z_{i^\star}\rangle$ gives the claim. □

Remark 1 (Ranks 2..k do not agree, and this is a result, not a gap). For $k > 1$ the two orderings are *not* equal in the limit. Ranks 2..k of π_β order the patterns by proximity to $z_{i^\star}$, whereas $\pi_{\cos}$ orders them by proximity to q. The reduction therefore holds exactly at $k = 1$ and as a statement about the retrieved *set* only when q is close enough to $z_{i^\star}$ that the two neighbourhoods coincide.

This is not a technical caveat; it is the mechanism behind the empirical shortlist regression in §5.1. Descent improves Recall@1 while degrading Recall@5 (0.605 vs. cosine's 0.759) precisely because the converged state has forgotten the cue. Our implementation of the dense baseline therefore reads the top-k from q directly, which is both the standard definition and the configuration that `tests/test_reduction.py` verifies; a baseline reading ranks 2..k off the converged state would be a weaker baseline and would flatter the method.

A.3 Ties

If the maximiser is not unique, let $\mathcal{M} = \arg\max_i s_i$ with $|\mathcal{M}| = m \geq 2$. Then $p_i(\beta) \to 1/m$ for $i \in \mathcal{M}$ and 0 otherwise, so $\boldsymbol{\xi}(\beta) \to \frac{1}{m}\sum_{i\in\mathcal{M}} z_i$, the centroid of the tied patterns, and Π maps it to the normalised centroid provided that sum is non-zero. The retrieved top-1 is then whichever tied pattern is nearest that centroid, which is an arbitrary tie-break — exactly as it is for exact cosine retrieval.

We record this case because it is the low-β centroid-collapse failure of §3.4 in miniature. At finite, small β the softmax is flat not because of exact ties but because $\beta\delta \ll 1$ for many pairs, and the state lands near the centroid of a large effective set. At β=16 on a 768-dimensional sentence encoder the effective set size $\exp(H(p))$ is 377 patterns. The tie case and the collapse case are the same phenomenon at different scales, which is why a single lower bound on β describes both.

A.4 Proof of Corollary 1

Each z_i is a fixed point of the update in the $\beta \to \infty$ limit: if $\boldsymbol{\xi}^{(t)} = z_j$ then $s = X^\top z_j$ is maximised uniquely at j (patterns distinct, unit norm), so by Lemma 2 the next state is z_j again. Combined with Lemma 2 applied at $t = 0$, the trajectory reaches $z_{i^\star}$ after one step and remains there, so $T > 1$ is observationally equivalent to $T = 1$, which is cosine retrieval by Proposition 1.

Scope of the empirical claim. Corollary 1 is exact only in the limit. The empirical statement in §3.4 — that a comparison run at high β compares cosine against cosine — is therefore an *approximation claim*, and we state its evidential basis precisely rather than leaning on the theorem. On held-out image completion at β=512, descent returns *the same memory as cosine on* 99.4% *of queries* (Recall@1 0.4986 against the cosine baseline's 0.5000). Decision agreement is the right statistic and accuracy is not: two retrievers can score identically while disagreeing about what they return, and the corollary is a claim about the choice. The theorem explains *why* the gap closes and gives the rate at which it closes; the measurement establishes that at the β values this literature actually uses, it has closed. Neither alone supports the claim.

A.5 Numerical verification

`tests/test_reduction.py` instantiates Equation (2) with T=1, β=10^6, b=0, η=0 and asserts top-1 agreement with an independent exact-cosine implementation, in two settings. On isotropic synthetic Gaussians (d=64, N=500) it agrees on 1000/1000 random queries. On real sentence embeddings (d=768, `all-mpnet-base-v2`) it agrees on 838/838 perturbed-episode queries.

The second setting is the one that carries weight, and we separate them deliberately. Random high-dimensional vectors are near-orthogonal, so the top-two gap δ of Lemma 1 is large and concentration is trivial. Real embeddings are anisotropic — unrelated pairs sit at cosine 0.258, not 0 — so δ is substantially smaller and the reduction is a materially stronger claim there. Both runs write their agreement counts to `experiments/results/reduction_check.json`; the test gates continuous integration on every commit. We use β=10^6 rather than a symbolic limit because that is the operating point the claim is about; Lemma 1 shows $\beta\delta \gg \log N$ is the sufficient condition and for $N \leq 10^4$, $\delta \geq 10^{-3}$ it is met with large margin.

B Hyperparameter grids, splits, and tuning budget

Table 9: Every grid searched, where it was searched, and where the number was reported. "Budget" is the number of configurations evaluated. Baselines receive a budget at least as large as the method they are compared against, which is the commitment §4 makes.

Experiment	Parameter grid	Budget	Selected on	Reported on
Image completion	$\beta \in \{12, \ldots, 512\}$, 14 values	14	tune half (images)	held-out half
	cosine: none	—	—	held-out half
Multi-β	scale sets, 6 conditions	6	tune half	held-out half
LoCoMo anchored	`anchor` $\times$ pp_0, 8 conditions	8	—	all 1,977 questions
	BM25: k_1, b grid	9	dev questions	test questions
	hybrid RRF: k constant	5	dev questions	test questions
Readout variants	$\beta \in \{16, 32, 64, 128\}$	4	dev storylines	test storylines
Association (ours)	$k_P \in \{5, 10, 20\}$, $\tau \in \{2, 5, 10\}$	9	val storylines, per fold	test storylines
	mixture K, β, α	48	val storylines, per fold	test storylines
Association (LoCoMo)	k_P, τ as above	9	val sessions, per fold	test sessions
	mixture K, β, α	18	val sessions, per fold	test sessions

The β grid in row 1 is swept densely through the transition region and coarsely outside it, and `sweep_beta_transition.py` records, for every value, both Recall@1 and the top-1 decision agreement with cosine plotted in Figure 1 (`experiments/results/beta_band.json`).

Cost. Measured wall-clock on the hardware of Appendix G, per query, on the held-out image split: cosine 0.0024 ms (1 matmul), single β at T=8 0.0502 ms (16 matmuls), integrated multi-β mixture 0.0810 ms (16 matmuls), and the m-way ensemble 0.1434 ms (48 matmuls). The ratio that matters for deployment is the first against the second: a 21× wall-clock cost for +0.013 Recall@1, in the one regime where descent wins. The mixture variants add a further 1.6–2.9× for an effect indistinguishable from zero (Table 6).

We report matmul counts alongside wall-clock because the two diverge: the integrated mixture and the single-β run issue the same 16 matmuls but differ 1.7× in wall-clock, since mixing m softmaxes per step is $O(mN)$ arithmetic outside the matmuls – 1.6× the wall-clock for the same matmul count. A matmul-only accounting would understate it.

C Corpus construction

The episodic corpus is generated by `src/energymem/eval/corpus.py`, deterministically from a seed. It contains 122 storylines drawn from 11 topic families (property, travel, health, gardening, vehicle, employment, study, finance, pets, home repair, social), each a chronological sequence of 3–8 episodes, interleaved in time so that the store is not grouped by topic. Multi-target questions are attached to storylines with the gold episode indices recorded at generation time.

Two deliberate difficulty devices:

Near-miss distractors. Each storyline has a planted counterpart in the same topic family that shares vocabulary but not referent — the apartment-hunting thread and the cottage thread both discuss commute times and deposits. These exist so that lexical overlap is an active hazard rather than an accident.

Compositional filler. 900 additional episodes generated from a subject/verb/time template grammar, which inflate the candidate set without introducing new storylines.

The X16 failure cases, in full. These are the clearest evidence in the paper that succession is not similarity, and they are real retrievals from the run, not illustrations. Every returned episode is a reasonable answer to "what is most similar?" and a wrong answer to "what happened next?"

Table 10: Cosine's top-1 for three one-hop association probes.

Cue	Gold (next episode)	Cosine returned
"It turns out to be a forty minute drive from any shop."	"Took most of the food with us and it was fine."	"Decided against Warren Street because the commute would be forty minutes each way." *(matched on "forty minute"; a flat-hunting episode)*
"Planted a crab apple at the end of the bed in October."	"First proper crop this autumn, so the theory held."	"Signed up for the weekly vegetable box." *(matched on growing/vegetables; different storyline)*
"Considering Iceland in February for the northern lights."	"Read that the ring road is often closed that time of year."	"Looking at the fjords in summer rather than winter." *(matched on Nordic travel; the Norway thread)*

Limitation, restated here because this is where a reviewer will look. This corpus is ours. It is generated, not collected, and §8 treats its external validity as unestablished. Appendix E reports the same probe design on a public benchmark built by another group, which is the only evidence that bears on whether the corpus was written to suit the answer.

D The experiment ledger

§7 makes claims about our own process—that errors were caught, that results were overturned by reimplementation, that nothing was quietly rewritten. Those claims are not checkable without the record, so the record ships with the paper.

`docs/10_EXPERIMENT_LEDGER.md` is append-only. Each entry carries an identifier, a date, the question asked, the result, and what the result changed. **An entry is never edited after its result is in.** When a later run overturns an earlier one, that is a new entry pointing back at the old, and the old entry stays as written—so X19 still records the +0.185 two-hop figure in its original wording, and X26b is the entry that withdraws it. A reader can therefore reconstruct not only what we believe now but what we believed then and why we stopped.

The ledger currently runs X1–X26c across five phases: synthetic validation (X1–X6), multi-answer readout (X7–X9), real embeddings (X10–X15), the world-model turn (X16–X19), the mode-averaging gate (X23–X25b), and external validity (X26–X26c).

One gap, stated rather than papered over. Three studies predate the ledger convention and carry no X identifier: image completion, the anchored LoCoMo sweep, and multi-scale β. They are cited in this paper by script name instead. We have deliberately not back-numbered them, because inserting entries into an append-only record retrospectively is precisely the manipulation the record exists to prevent. The ledger preamble says the same.

E LoCoMo association probes

Task. Cue is a conversational turn; the gold target is the turn that followed it in the same session. This is the same probe design as the episodic corpus (Appendix C), applied to a benchmark we did not build.

Candidate set. The *full pooled bank* of all 5,882 turns from all ten conversations. A prediction must beat distractors from nine unrelated conversations as well as from its own, which makes the task strictly harder than per-conversation retrieval and is why absolute scores are lower here than on the episodic corpus (cosine Recall@1 0.064 vs. 0.105).

Namespacing. Dialogue identifiers are namespaced by `sample_id`. Without this, 5,882 turns carry only 1,033 distinct identifiers and a turn from an unrelated conversation satisfies a gold identifier and scores as a hit — the bug that invalidated an earlier capacity sweep of ours (§5.2).

Split unit. The *session* (272 of them across ten conversations), not the conversation and not the turn. A session is a coherent conversational episode; transitions from a session the model was fitted on are never evaluated. Four-fold rotation places every session in the test set exactly once, with the validation fold disjoint from both. The world model is fitted on ≈2,800 train-session transitions per fold, and both (k_P, τ) and the readout's (K, β, α) are selected on that fold's validation sessions.

Probe subsampling. Sessions vary in length, so at most 25 probes per session enter the evaluation, sampled with a fixed seed. This keeps folds balanced and costs nothing statistically: the bootstrap resamples sessions, so the interval is driven by the 272 clusters rather than by the probe count.

Full results. Table 11. Note the asymmetry that bounds the claim: at one hop the effect is large and significant on both metrics; at two hops it is null at Recall@1 and carries only a residual +0.012 at Recall@5, an order of magnitude smaller.

Table 11: LoCoMo association, all methods and both hops. Hop 1 n=5,233; hop 2 n=5,013; 272 session clusters throughout; intervals are 95% cluster bootstrap. `experiments/locomo_assoc.py`; raw values in `experiments/results/x26_locomo_assoc.json`.

Hop	Method	R@1	Δ R@1	R@5	Δ R@5
1	cosine	0.064	—	0.174	—
	averaged	**0.107**	+0.043 [+0.036, +0.050]	**0.251**	+0.077 [+0.067, +0.087]
	mixture	0.075	+0.011 [+0.006, +0.016]	0.202	+0.028 [+0.022, +0.035]
2	cosine	0.043	—	0.127	—
	averaged	0.041	−0.002 [-0.008, +0.004]	0.139	+0.012 [+0.004, +0.019]
	mixture	0.045	+0.002 [-0.003, +0.006]	0.134	+0.007 [+0.002, +0.012]

The mixture readout, on public data. §5.5 reports that preserving the predictor's modes loses to averaging them on our corpus. It replicates here in sign and magnitude: the mixture beats cosine (+0.011) and loses decisively to averaging (0.075 vs. 0.107, a gap of −0.032). Invariant I4 is falsified on both corpora.

F Statements

Code and data availability. All code, the append-only experiment ledger, and every results file behind every table and figure are released under an open licence at the repository accompanying this paper. Neither evaluation corpus is redistributed: both are public and are fetched by the two commands in Appendix G. The third corpus is generated deterministically from a seed by code in the repository rather than stored, so the exact queries are reproducible without a data file.

Reproducibility. Every table and figure with a results file behind it is regenerated by one command (Appendix G), and the generator prints on every run any table it could not regenerate. Proposition 1 is asserted in continuous integration and writes its agreement counts to a results file.

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Ethics. This work uses two public benchmark corpora and a synthetic corpus of the authors' own construction. It involves no human subjects, no personal data, and no deployment. The agent-memory setting it studies is dual-use only in the ordinary sense that better retrieval improves any system built on it; we identify no specific harm pathway beyond that.

G Reproducibility

Environment. All results were produced on a single machine: Apple M5 Max, 128 GB RAM, macOS 26.5.1 (arm64). Python 3.13.2, NumPy 2.5.2, PyTorch 2.14.0, torchvision 0.29.1, sentence-transformers 6.0.1.

No GPU-specific code paths; every experiment runs on CPU. Encoders are frozen pretrained checkpoints (`all-mpnet-base-v2`; `ResNet18_Weights.IMAGENET1K_V1`) and are never fine-tuned.

Data. Neither corpus is redistributed. LoCoMo (2.8 MB) and Imagenette-160 (99 MB) are fetched by the two `curl` commands in the repository README. The episodic corpus is *generated*, not stored: it is produced deterministically from a seed by `src/energymem/eval/corpus.py`, so the exact queries are reproducible without shipping a data file.

Regenerating the paper. Every table and figure that has a results file behind it is produced by one command, and tables are written to `paper/generated/` and `\input`ed rather than typed:

```
make setup
pytest -q                                   # Prop. 1 gate; writes reduction_check.json
python experiments/x23_mode_diagnostics.py  # X23-X25
python experiments/locomo_assoc.py          # X26
python scripts/exp_locomo_anchored.py       # Table 2
python scripts/confirm_holdout.py           # Table 1
python scripts/exp_multibeta_scales.py      # Table 5
python scripts/make_tables.py               # JSON -> generated tables + figures
cd paper && pdflatex main && bibtex main && pdflatex main && pdflatex main
```

`make_tables.py` prints, on every run, the list of tables it could *not* generate from a results file. We regard that output as part of the reproducibility statement: it names precisely which numbers in the paper are still typed rather than derived, so the gap is visible to a reader instead of assumed closed.

Known reproduction frictions. Two, both environmental rather than scientific, recorded because they cost us time. Python on this machine ships without a CA bundle, so `urllib`-based downloads (including `torch.hub` weight fetches) fail with a certificate error; export `SSL_CERT_FILE` from `certifi`, or pre-fetch the checkpoint with `curl`. And `torchvision` is an optional dependency not installed by `make setup`; the two image experiments require it.

Every number currently reported in this paper is reproducible. Two drivers were lost before they were committed — the one behind Table 3 row (i) and the one behind the parametric-predictor result in §5.3. Both have been rewritten — `x17b_single_split.py` and `x17_parametric.py` under `experiments/` — because the modules themselves survived and only the drivers were missing. No figure in this paper now depends on code nobody can run.

Rewriting them was not free of consequence, and the consequence is reported rather than buried. The parametric result reproduced more starkly than recorded (fitted Recall@1 1.000 and unseen 0.000 at rank 32, against the original 0.784 and 0.008) and the paper cites the re-run. Row (i) did *not* reproduce: the re-run recovers the cosine baseline exactly but gives a non-significant +0.024 at n=123 where the original recorded a significant +0.049 at n=37. The table reports the re-run and marks it n.s.; §5.3 says why the claim does not depend on it. This is the sixth entry in the error taxonomy of §7 and the sharpest form of its argument: **an uncommitted script is not a result.**